International Journal on Science and Technology (IJSAT)
E-ISSN: 2229-7677 ● Website: www.ijsat.org ● Email: editor@ijsat.org

Quantum Machine Learning for Financial Forecasting and Portfolio Optimization: Algorithms, Applications, and Future Prospects

Srinivasa Kalyan Vangibhurathachhi

Srinivasa.Kalyan2627@gmail.com

Abstract:
Within the current complex, volatile, and data-intensive finance sector, quantum machine learning is emerging as a critical tool in financial forecasting and portfolio optimisation. Major financial players are leveraging quantum computing's ability to process complex, high-dimensional data, as quantum machine learning algorithms offer significant improvements over classical methods in terms of speed, accuracy, and scalability. This article evaluates key quantum machine learning models, including quantum support vector machines, variational circuits, and quantum annealing, while also examining their sector-specific applications in forecasting and asset allocation. It also addresses current limitations inhibiting broader adoption of quantum machine learning, such as hardware immaturity and regulatory gaps, while highlighting emerging solutions.

Keywords: Quantum Machine Learning, Financial Forecasting, Portfolio Optimisation, Quantum Algorithms, Hybrid Quantum-Classical Systems.

1. INTRODUCTION

In today's digital era, financial markets are inundated with unprecedented volumes of data whose complexity and volatility are rising. Traditional machine learning and classical computing methods are unable to handle the sheer scale and uncertainty inherent in financial forecasting and portfolio optimisation (Bhasin et al., 2024; Zhou, 2025). To respond to this challenge, there is a growing interest in Quantum Machine Learning (QML), the convergence of quantum computing and machine learning, within the financial industry as a critical tool for financial forecasting and portfolio optimisation (Patel, 2025; Vashishth et al., 2025). This has resulted in a 33.8% compound annual growth rate of the Quantum Machine Learning market from $1.12 billion in 2024 to $1.5 billion in 2025 (Business Research Company, 2025). Quantum computing in the financial services market alone is valued at $0.3 billion and is expected to reach $6.3 billion by 2023, which represents a 46.5% growth rate. This indicates the interest in quantum computing culminating in real investment in the technology.

Major financial institutions are already investing and leveraging the technologies. JPMorgan Chase has invested $100 million in quantum computing company Quantinuum while other financial institutions like Wells Fargo are working with IBM to create nearly a dozen quantum algorithms (Crosman, 2024). Similarly, Goldman Sachs researchers have been working with Amazon Web Services to assess the practicality of quantum algorithms (Crosman, 2024). A 2024 report by Moody's review indicated that the financial sector stands to benefit the most from the quantum applications as a result of the high-complexity nature of its use cases. According to the Quantum Zeitgeist (2025), 80% of the major global financial institutions are engaging with quantum computing, with the impact reaching $622 billion by 2025 due to enhanced portfolio optimisation as well as fraud detection.

These trends accentuate quantum machine learning's potential to revolutionise the financial market as it offers faster training time, enhanced ability to handle volatile data, and transformative efficiency in optimisation. This article explores how quantum algorithms, including quantum neural networks and

variational circuits, are reshaping financial forecasting and portfolio management while also identifying the critical challenges to broader adoption. The specific objectives of the articles are:

i. Evaluate the core quantum-machine-learning algorithms used in financial markets.
ii. Analyse sector-specific applications of quantum machine learning in finance.
iii. Identify the limitations and challenges hindering broader adoption of quantum machine learning in finance.
iv. Examine emerging solutions and the future outlook for quantum machine learning in financial markets.

2 PROBLEM STATEMENT

Financial markets rely on predictive models to optimise portfolios, manage risks, and enhance forecasts. Yet as the volume of financial data surges and volatility and complexity become defining traits of the market, traditional computational approaches, such as Monte Carlo simulations, linear regression, and classical support-vector machines, prove increasingly insufficient and ineffective (Deep, 2024; Behera et al., 2024). For instance, the Classical Monte Carlo simulations require extensive computational time when modelling complex derivatives or even assessing value-at-risk in multi-factor models. This makes them inefficient for facilitating real-time decision-making, which is necessary within the current volatile and fast-moving markets (Alonso, 2025). This has created a critical need for enhanced computational paradigms. Quantum machine learning, which merges quantum computing and machine learning frameworks, offers a compelling solution to this challenge. Quantum machine learning promises exponential speedups and superior generalisation in big data and complex environments, which makes them ideal for tackling the computational and structural challenges of modern finance.

3.0 CORE QUANTUM MACHINE LEARNING ALGORITHMS

3.1 Quantum Support Vector Machines

Quantum support vector machine is a quantum analogue of classical support vector machine, which is usually utilised for classifying tasks. This is useful in credit risk assessment as well as market regime detection (Akrom, 2024). Quantum Support Vector Machines leverages quantum feature mapping. Which allows the embedding classical data into a high-dimensional Hilbert space, where data patterns are linearly separable (Slabbert and Petruccione, 2024). This enables faster computation of inner products between data points, which traditionally bottlenecks classical Support Vector Machines as the dataset size increases (Slabbert and Petruccione, 2024). Quantum Support Vector Machines has demonstrated success in early experiments using IBM's quantum devices for binary classification tasks. In financial applications, these systems are able to distinguish between bullish and bearish markets or classic assets by risk profiles (Doosti et al., 2024). Quantum Support Vector Machines shows massive potential for efficiently handling large, non-linearly separable datasets in finance, potentially reducing runtime from hours to minutes for different tasks.

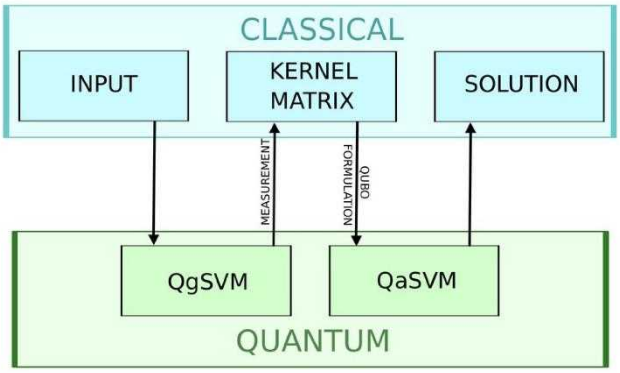

Figure 1: Quantum Support Vector Machines

3.2 Variational Quantum Circuits

Variational Quantum Circuits are hybrid quantum-classical models used primarily for regression and supervised learning problems, such as price forecasting, asset valuation, and yield curve modelling (Khurana and Nene, 2025). A variational quantum circuit has a parameterised quantum circuit trained via a classical optimiser to minimise a cost function, enabling it to learn complex and non-linear relationships in data (Li and Jiang, 2024). Variational Quantum Circuits are preferable in noisy quantum environments and are compatible with existing Noisy Intermediate-Scale Quantum devices. This represents richer functional forms in comparison to traditional neural networks due to quantum superposition and entanglement, which encode correlations more efficiently (Dutta and Sandeep, 2025). As such, financial institutions such as Goldman Sachs and QC Ware have explored VQCs for volatility prediction and structured product pricing (QC Ware, 2022). These circuits offer a more expressive and compact way to model complex market dynamics than many traditional deep learning architectures.

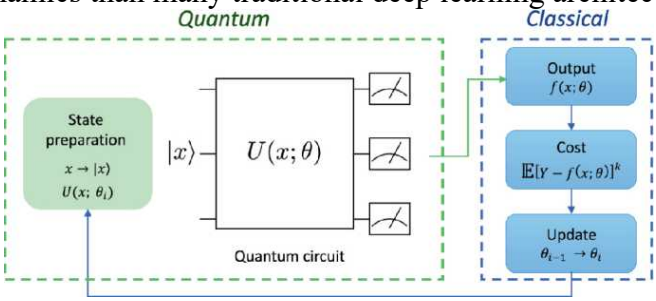

Figure 2: Variational Circuit Based Hybrid Quantum-Classical Algorithm

3.3 Quantum Principal Component Analysis

Quantum Principal Component Analysis reduces the dimensionality of large datasets, which is a critical task in financial markets analytics, as thousands of variables need to be analysed in real-time (Patel, 2025). Classical Principal Component Analysis have significant computational bottlenecks when calculating covariance matrices and eigenvalue of a matrix on massive data streams (Auddy et al., 2024) In contrast, Quantum Principal Component Analysis uses quantum phase estimation and density matrix involution to identify the principal components of a dataset exponentially faster than classical Principal Component Analysis in theory (Sihare, 2025). This enables efficient compression of portfolio data, macroeconomic indicators, and trading signals, enhancing performance in downstream tasks like asset clustering or risk decomposition. Quantum Principal Component Analysis shows massive potential for managing complex high-dimensional datasets that overwhelm classical systems, particularly in high-frequency trading and market microstructure modelling where dimensionality reduction is essential.

3.4 Quantum Annealing for Portfolio Optimisation

Quantum annealing is a meta-heuristic technique that uses quantum tunnelling to find optimal points in optimisation landscapes. This makes it effective in solving difficult problems in financial analytics such as portfolio optimisation (Tripathi et al., 2025). The traditional portfolio construction methods like Markowitz's mean-variance framework need approximations when dealing with large, constrained asset universes, often leading to suboptimal allocations due to computational limits (Hao, 2025). Quantum annealers can encode portfolio constraints, such as budget limits and sector exposure, into a quadratic unconstrained binary optimisation problem, which enables the system to evaluate multiple configurations simultaneously and escape false bottom (local minima) more effectively than simulated annealing. According to Quinton et al. (2025), quantum annealing produces comparable or better optimisation outcomes in a fraction of the time required by classical solvers. As quantum hardware scales, this approach is likely to become a critical tool for asset managers and hedge funds seeking real-time rebalancing under complex constraints.

4.0 SECTOR-SPECIFIC APPLICATIONS

4.1 Financial Forecasting

Financial managers need to predict market movements, asset prices, volatility, and macroeconomic indicators within the complex and fast-moving world. However, the available traditional models, like the classical neural networks, have limited abilities within environments of high volatility and big data (Han et al., 2024). This has resulted in consideration of quantum machine learning approaches, as they are more advanced and can capture intricate temporal and non-linear relationships within large relationships within massive datasets more efficiently (Alcazar et al., 2020). For instance, as presented by Quantum Zeitgeist (2024), JPMorgan is already utilising quantum-powered Monte Carlo simulations to predict future asset price distributions and has already demonstrated that quantum machine learning reduces computational time for risk modelling significantly. In addition, hybrid quantum-classical models have demonstrated improved accuracy in predicting financial market volatility indices as well as yield curves, outperforming traditional models (Alcazar et al., 2020). The ability of quantum machine learning approaches to process high-dimensional data simultaneously allows for improved scenario analysis that considers all factors, which facilitates more efficient anomaly detection. With the growing complexity of financial systems, quantum-based forecasting presents a transformative tool for central banks, investment firms, and hedge funds to anticipate trends, detect and mitigate risks at improved speed and precision.

4.2 Portfolio Optimisation

Portfolio optimisation is the strategic balancing of investment to attain optimum performance while managing uncertainty, often considering liquidity, diversification, and regulatory requirements (Rodrigues et al., 2025). Classical approaches of portfolio optimisation that rely on convex optimisation, such as Markowitz's mean-variance optimisation, are limited when handling real-world constraints such as cardinality limits, transaction costs and well as non-linear risk factors (Lindell and Hultqvist, 2025). Quantum algorithm, such as quantum approximate optimisation algorithms and quantum annealing, solves these limitations as they are able to factor in non-linear factors as well as real-world constraints effectively (Torta, 2024; Venturelli and Kondratyev, 2019). These techniques offer solutions after considering complex issues and a massive amount of non-linear data by utilising quantum tunnelling to identify optimal allocations that classical approaches have a challenge attaining. As presented by Mazouni (2024), private equity firms as well as venture capital firms are already exploring quantum portfolio models to optimise asset allocation in real-time. These institutions are using D-Wave's quantum annealers to optimise their portfolios. By enabling enhanced scenario analysis and faster portfolio rebalancing, quantum optimisation will become a critical and mainstream asset for institutional investors optimising multi-asset portfolios under uncertainty as well as in high-frequency trading where real-time actions are critical.

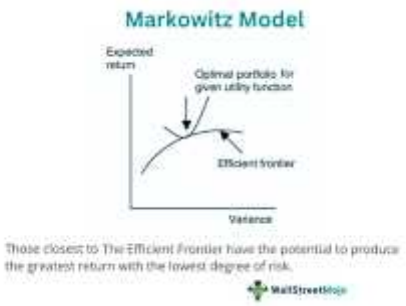

Figure 3: Markowitz's mean-variance optimization

5.0 CHALLENGES TO ADOPTION

5.1 Hardware Constraints

Current quantum computers are limited by high error rates, high cost per qubit counts, and short coherence times (Yang et al., 2023). These are significant constraints that limit the complexity and depth of quantum machine learning models that can be executed effectively and that offer reliable solutions. The high cost

of hardware also makes broader adoption challenging, given key players within the financial sector do not consider it a mature technology. Hence, until the issues of errors are addressed and large-scale quantum systems are mainstreamed, quantum machine learning applications will continue to depend on hybrid classical-quantum that will only partially boost workflow efficiency rather than offer the full advantages of quantum.

5.2 Algorithmic Barriers and Regulatory Considerations

Quantum machine learning is still at the early stages of development; hence, it lacks standardised benchmarks and proven performance on real-world financial datasets. Designing an effective quantum circuit within the high-dimensional and noisy data financial system is complex. Unlike the classical machine learning methods that have been refined over time and which have community-validated heuristics, the quantum methods are experimental and untested. This presents massive algorithmic barriers to its application in real-world scenarios, which have massive in turns of financial risks as well as reputational risks. Furthermore, given that the technology is in its early phases, it's difficult to offer regulations other than requiring transparency and accountability.

6.0 FUTURE OUTLOOK

Quantum-inspired algorithms, which are classical algorithms that use quantum computing to solve problems efficiently compared to traditional methods, without requiring actual quantum hardware, will become a critical tool in financial markets going forward (Gharehchopogh, 2023). These algorithms utilise techniques such as tunnelling, entanglement, and superposition, without requiring actual quantum hardware (Mandal and Chakraborty, 2025). These approaches offer increased computational advantages for solving non-linear problems in finance. These techniques enhance portfolio optimisation outcome, achieving greater diversification and faster computation compared to traditional techniques. These algorithms present a bridge between full-scale quantum systems, offering an opportunity for firms to experiment with quantum principles before full-scale deployment of quantum machine learning.

Initially, quantum machine learning succeeded by utilising a hybrid architecture that integrates quantum models and classical financial systems. These integrations ensure regulatory compliance through auditable workflow while also boosting the computational speed in critical areas such as real-time optimisation and scenario simulation. Going forward, institutions will continue to embed quantum solvers within traditional data pipelines to enhance workflow efficiency.

7.0 Conclusion

Quantum Machine Learning presents massive potential for revolutionising the financial sector as its enhanced computational capabilities can solve problems previously constrained by classical computational limits. Quantum Machine Learning has demonstrated improved efficiency in forecasting, optimisation, and trading, as it offers speed and precision that massively boosts financial analytics and strategy. While some limitations hinder broader adoption, such as hardware and regulation, advancements in hybrid architectures and quantum-inspired methods are already enabling real-world integration where companies are utilising quantum algorithms and classical methods. To leverage the full benefits of Quantum Machine Learning's potential, financial institutions need to invest in talent, infrastructure, and ethical governance. The growing adoption of quantum in finance marks a major technological leap and also strategic transformation toward a more agile, efficient, and adaptive financial ecosystem.

REFERENCES:

[1] Akrom, M. (2024). Quantum support vector machine for classification task: A review. Journal of Multiscale Materials Informatics, 1(2), 1-8.

[2] Alcazar, J., Leyton-Ortega, V., & Perdomo-Ortiz, A. (2020). Classical versus quantum models in machine learning: insights from a finance application. Machine Learning: Science and Technology, 1(3), 035003.

[3] Alonso, N. I. (2025). Numerical Methods in Quantitative Finance. Numerical Methods in Quantitative Finance
[4] Auddy, A., Xia, D., & Yuan, M. (2024). Tensor methods in high dimensional data analysis: Opportunities and challenges. arXiv preprint arXiv:2405.18412.
[5] Behera, I., Nanda, P., Mitra, S., & Kumari, S. (2024). Machine Learning Approaches for Forecasting Financial Market Volatility. Machine Learning Approaches in Financial Analytics, 431-451.
[6] Bhasin, N. K., Kadyan, S., Santosh, K., HP, R., Changala, R., & Bala, B. K. (2024, March). Enhancing quantum machine learning algorithms for optimized financial portfolio management. In 2024 Third International Conference on Intelligent Techniques in Control, Optimization and Signal Processing (INCOS) (pp. 1-7). IEEE.
[7] Business Research Company (2025). *Quantum Machine Learning Market Report 2025 - Quantum Machine Learning Market Forecast and Overview*. [online] Thebusinessresearchcompany.com. Available at: https://www.thebusinessresearchcompany.com/market-insights/quantum-machine-learning-market-overview-2025 [Accessed 21 Jul. 2025].
[8] Crosman, P. (2024). *Quantum leap: JPMorgan Chase, Wells Fargo push past laggard banks*. [online] American Banker. Available at: https://www.americanbanker.com/news/quantum-leap-jpmorgan-chase-wells-fargo-push-ahead-while-most-banks-lag.
[9] Deep, A. T. (2024). Advanced financial market forecasting: integrating Monte Carlo simulations with ensemble Machine Learning models.
[10] Doosti, M., Wallden, P., Hamill, C. B., Hankache, R., Brown, O. T., & Heunen, C. (2024). A brief review of quantum machine learning for financial services. arXiv preprint arXiv:2407.12618.
[11] Dutta, S. S., & Sandeep, S. (2025). Hybrid quantum neural networks: harnessing dressed quantum circuits for enhanced tsunami prediction via earthquake data fusion. EPJ Quantum Technology, 12(1), 4.
[12] Gharehchopogh, F. S. (2023). Quantum-inspired metaheuristic algorithms: comprehensive survey and classification. Artificial Intelligence Review, 56(6), 5479-5543.
[13] Han, H., Liu, Z., Barrios Barrios, M., Li, J., Zeng, Z., Sarhan, N., & Awwad, E. M. (2024). Time series forecasting model for non-stationary series pattern extraction using deep learning and GARCH modeling. Journal of Cloud Computing, 13(1), 2.
[14] Hao, R. R. (2025). Novel Portfolio Designs Based on Markowitz Portfolio Theory and Various Assets.
[15] Khurana, S., & J Nene, M. (2025). Integration of parameterized quantum circuits within classical neural network for financial time-series prediction. Quantum Machine Intelligence, 7(1), 36.
[16] Li, J., Li, A., & Jiang, W. (2024, April). Quapprox: A framework for benchmarking the approximability of variational quantum circuit. In ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 13376-13380). IEEE.
[17] Lindell, M., & Hultqvist, A. (2025). Portfolio Optimisation In Practice: Risk and Return: A Comparative Out-of-Sample Analysis of Portfolio Optimisation Methods: Evidence from the Swedish Equity Market.
[18] Mandal, A. K., & Chakraborty, B. (2025). Quantum computing and quantum-inspired techniques for feature subset selection: A review. Knowledge and Information Systems, 67(3), 2019-2061.
[19] Mazouni, M. (2024). Quantum-Inspired Private Equity and Venture Capital Fund Optimization.
[20] Moodys (2025). *Quantum Computing in the Financial Sector: 2024 Trends in Review*. [online] Moodys.com. Available at: https://www.moodys.com/web/en/us/insights/quantum/quantum-computing-in-the-financial-sector-2024-trends-in-review.html
[21] Patel, A. (2025). ADVANCING FINANCIAL PREDICTION THROUGH QUANTUM MACHINE LEARNING. International Journal of Intelligent Data and Machine Learning, 2(02), 1-7.
[22] Patel, A. (2025). ADVANCING FINANCIAL PREDICTION THROUGH QUANTUM MACHINE LEARNING. International Journal of Intelligent Data and Machine Learning, 2(02), 1-7.

[23] QC Ware (2022). *QC Ware*. [online] Qcware.com. Available at: https://www.qcware.com/news/goldman-sachs-and-qc-ware-collaboration-brings-new-way-to-price-risky-assets-within-reach-of-quantum-computers.
[24] Quantum Zeitgeist. (2024). *Quantum Computing And Financial Modeling The Future Of Finance*. [online] Available at: https://quantumzeitgeist.com/quantum-computing-and-financial-modeling-the-future-of-finance/.
[25] Quantum Zeitgeist. (2025). *Evident Research Assesses Quantum Readiness In Major Global Banks*. [online] Available at: https://quantumzeitgeist.com/evident-research-assesses-quantum-readiness-in-major-global-banks/.
[26] Quinton, F. A., Myhr, P. A. S., Barani, M., Crespo del Granado, P., & Zhang, H. (2025). Quantum annealing applications, challenges and limitations for optimisation problems compared to classical solvers. Scientific Reports, 15(1), 12733.
[27] Rodrigues Coelho, F. I., Bizarrias, F. S., Rabechini Jr, R., Martens, C. D. P., & Martens, M. L. (2025). Strategic alignment and value optimization: Unveiling the critical role of project portfolio management for a flexible environment. Global Journal of Flexible Systems Management, 26(1), 209-224.
[28] Sihare, S. R. (2025). Dimensionality Reduction for Data Analysis With Quantum Feature Learning. Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery, 15(1), e1568.
[29] Slabbert, D., & Petruccione, F. (2024). Hybrid Quantum-Classical Feature Extraction approach for Image Classification using Autoencoders and Quantum SVMs. arXiv preprint arXiv:2410.18814.
[30] Torta, P. (2024). Quantum Approximate Optimization Algorithm and Variational Quantum Computing: from binary neural networks to ground state preparation.
[31] Tripathi, R., Tomar, S., & Kumar, S. (2025). A Comprehensive Survey on Quantum Annealing: Applications, Challenges, and Future Research Directions. Authorea Preprints.
[32] Vashishth, T. K., Sharma, V., Kaushik, V., Singh, A., & Kaushik, S. (2025). Exploring quantum challenges and opportunities of quantum machine learning adoption in finance. Machine Learning and Modeling Techniques in Financial Data Science, 437-460.
[33] Venturelli, D., & Kondratyev, A. (2019). Reverse quantum annealing approach to portfolio optimization problems. Quantum Machine Intelligence, 1(1), 17-30.
[34] Yang, Z., Zolanvari, M., & Jain, R. (2023). A survey of important issues in quantum computing and communications. IEEE Communications Surveys & Tutorials, 25(2), 1059-1094.
[35] Zhou, J. (2025). Quantum finance: Exploring the implications of quantum computing on financial models. Computational Economics, 1-30.